

uncertainty. There are two key drivers for the selection of the appropriate optimisation method: statistical data availability and desired degree of robustness. Robust optimisation has emerged as a leading method for problems with limited uncertain parameter data and requirements for a risk-averse solution. In a robust optimisation context, solutions that lead to attainable decisions under all parameter realisations are referred to as robust feasible, while the solution that also leads to the best objective value is called robust optimal. A general form of the robust optimisation problem is given by (P_{rob}) .

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & g(x, u) \leq 0, \quad \forall u \in \mathcal{U} \end{aligned} \tag{P_{rob}}$$

where $x \in \mathcal{X} \subseteq \mathbb{R}^{n_x}$ denotes continuous decisions, and u is the vector of uncertain parameters that resides within an uncertainty set $\mathcal{U}$. Despite the presence of an embedded optimisation problem within (P_{rob}) , robust optimisation problems can yield tractable reformulations under restrictive assumptions about the problem structure. That is, if Problem (P_{rob}) is convex with respect to the decisions x and concave in u , and $\mathcal{U} \subset \mathbb{R}^{n_u}$ is a convex set, then using duality theory one can derive a tractable robust counterpart reformulation of Problem (P_{rob}) . Because of this, the field of convex robust optimisation is well-established, whereas the field of non-convex robust optimisation is still at its infancy [1].

State-of-the-art robust optimisation algorithms for non-convex problems typically rely on global solvers to obtain robust optimal solutions. The prevailing approach involves deriving the dual reformulation and utilising a global solver, either directly or adaptively. However, the dual reformulation can increase the problem complexity, and in the case of challenging non-convex problems may only yield a robust feasible solution. An alternative approach is based on an iterative robust cutting plane algorithm. In this case, significant computational time can be spent searching for a global solution that may be deemed robust infeasible.

Contributions of this work: Our research hypothesis is that integrated exploration of global and robust optimality can yield computational benefits. To this end, in this article we introduce a novel algorithm that conducts a concurrent global optimality and robustness search for continuous non-convex problems. The proposed approach integrates spatial branch-and-bound with robust cutting plane notions. The key idea is that, while exploring the branch-and-bound tree, we assess the robustness of nodes entailing the best-found solutions. We illustrate the performance and benefits of our proposed approach through benchmark Quadratically Constrained Quadratic Programs (QCQPs) of pooling problems. At each node, the non-convex problem is solved via a local solver. If the computed solution is as good as the best-found so far, an infeasibility test is performed to evaluate the robustness of the obtained solution. If not, then the corresponding cutting planes are added both to the non-convex and convex problems. The algorithm proceeds to the next step once no more violations are detected. The convex problem is solved next, and the solution of this problem is used to decide the most promising variable for branching into two child nodes. With the use of appropriate fathoming criteria, the tree nodes are exhausted, and the robust optimal solution is obtained.